

# Anirudh Som

## Contact Info:

**Email:** [contact.anirudh.som@gmail.com](mailto:contact.anirudh.som@gmail.com) **Website** [Website](#) **LinkedIn** [LinkedIn](#) **Google Scholar** [Google Scholar](#)

**Work Authorization:** U.S. Permanent Resident

## Skills:

**Programming** – Python, Matlab, XML, Latex

**Libraries & Frameworks** – PyTorch, TensorFlow, Transformers, Scikit-learn, Pandas, OpenCV, DeepSpeed

**LLMs & VLMs** – GPT, LLaMa, Mixtral, Mistral, Open LLaMa | RAG, RLHF, SFT, LoRA, PEFT, In-Context Learning

**Evaluation Benchmarks** – HELM, MMLU, TruthfulQA, BLEU, ROUGE, BERTScore, CIDEr

**Infrastructure** – Docker, Kubernetes, Gitlab, Artifactory, AWS, GPU cluster training, FSDP, ZeRO

**Data Handling** – Document parsing, OCR pipelines, Structured extraction from unstructured sources

## Education:

**Ph.D., Electrical Engineering** – Arizona State University, Tempe, AZ

2020

## Professional Experience:

• **Vice President - Applied AI/ML** – JPMorganChase, New York, NY

2025 – Present

- o Part of the Payment Solutions team in the Commercial & Investment Banking unit.
- o Developing a payments-focused foundation model leveraging large-scale transactional and merchant data, with a focus on transferability across multiple downstream tasks.

• **Advanced Computer Scientist** – SRI, Princeton, NJ

2021 – 2025

- o **CV & NLP Systems:** Led development and deployment of transformer-based CV and LLM pipelines for real-time and offline video understanding, including scene segmentation, summarization, metadata extraction, and emotion-aware HMLs.
- o **Document Intelligence:** Built OCR+NLP pipelines and LLM systems to extract structured insights from unstructured medical and educational documents.
- o **Mentorship & Leadership:** Served as technical lead on federal and commercial AI projects, mentoring and supervising 2–4 junior ML engineers and interns.
- o **Commercial Use Cases:** Delivered AI prototypes and proof-of-concepts for Fortune 500 clients, including L'Oréal and Honda.
- o **Product Collaboration:** Collaborated with cross-functional teams to deploy and maintain ML solutions aligned with product goals and operational needs.

### Key Projects:

- o **ARPA-H PARADIGM:** CV+LLM-driven task guidance system for rural healthcare support.
- o **DARPA CCU:** CV+LLM pipelines for cross-cultural awareness in dialogue systems.
- o **DARPA EDGE:** Real-time emotion-aware HMI systems.
- o **NSF:** Automated classroom collaboration assessment using multimodal architectures.

## Pre-Doctoral Experience:

• **Intern** – SRI, Princeton, NJ

2020

Multimodal deep learning for educational assessment.

• **Intern** – Roche Diagnostics, Santa Clara, CA

2019

Image segmentation, federated learning in digital pathology.

• **Intern** – Lawrence Livermore National Laboratory, Livermore, CA

2018

Time-series deep learning for heart signal modeling.

• **Intern** – Mayo Clinic, Phoenix, AZ

2015

ML for cancer detection in CT/MRI.

## Selected Publications:

- [1] **A. Som** et al. "[MINDS: A Cross-cultural Dialogue Corpus for Social Norm Classification and Adherence Detection](#)". Accepted at the International Joint Conference on NLP & Asia-Pacific Chapter of the Association for Computational Linguistics (IJCNLP-AAACL) Findings, 2025.
- [2] **A. Som** et al. "[Demonstrations Are All You Need: Advancing Offensive Content Paraphrasing using In-Context Learning](#)". Accepted at the Association for Computational Linguistics (ACL) Findings, 2024.
- [3] **A. Som**, et al. "[Automated Student Group Collaboration Assessment and Recommendation System Using Individual Role and Behavioral Cues](#)". Accepted at the Frontiers in Computer Science Journal, 2021.
- [4] **A. Som**, et al. "[Perturbation Robust Representations of Topological Persistence Diagrams](#)". Accepted at the European Conference on Computer Vision (ECCV), 2018.

**Complete List:** [Google Scholar](#)

## Patents:

- [1] [Machine Learning Model Prompt Demonstration Selection.](#)
- [2] [Automated Identification of Necrotic Regions in Digital Images of Multiplex Immunofluorescence-Stained Tissue.](#)
- [3] [Federated Learning System for Training Machine Learning Algorithms Without Accessing Patient Data to Protect Patient Privacy.](#)
- [4] [Automated Collaboration Skills Assessment.](#)

## Awards:

- "Research Recognition Award" – CVT Group at SRI
- "Facilitators' Choice Award" - 2021 NSF STEM for All Video Showcase

2022 – 2024  
2021